

Real II

* Prove that $\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1$

$$\text{Let } a_n = \frac{1}{n(n+1)} \quad \forall n \in \mathbb{N}$$

$$\text{The } S_n = a_1 + a_2 + \dots + a_n = \sum_{k=1}^n a_k \quad \leftarrow n^{\text{th}} \text{ partial sum}$$

$$a_n = \frac{1}{n(n+1)}$$

$$a_n = \frac{1}{n} - \frac{1}{n+1}$$

$$a_1 = 1 - \frac{1}{2}$$

$$a_2 = \frac{1}{2} - \frac{1}{3}$$

$$a_3 = \frac{1}{3} - \frac{1}{4}$$

$$\vdots$$

$$a_n = \frac{1}{n} - \frac{1}{n+1}$$

$$\therefore a_1 + a_2 + a_3 + \dots + a_n = 1 - \frac{1}{n+1}$$

$$\sum_{k=1}^n a_k = 1 - \frac{1}{n+1}$$

$$\sum_{k=1}^n a_k = S_n$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n+1}\right) = 1 //$$

That is the sequence (S_n) converges to 1.

$$\therefore \sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1 //$$

* Consider $\sum_{n=1}^{\infty} 1$

Here $a_n = 1$

$$S_n = a_1 + a_2 + \dots + a_n = 1 + 1 + 1 + \dots + 1 = n \quad \leftarrow n^{\text{th}} \text{ partial sum.}$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} n = \infty //$$

$\therefore S_n$ is divergent and hence the infinite series,

$$\sum_{n=1}^{\infty} 1 \text{ is divergent.} //$$

* Consider $\sum_{n=1}^{\infty} n \sin(\frac{1}{n})$.

Let $a_n = n \sin(\frac{1}{n})$.

$$\begin{aligned} \lim_{n \rightarrow \infty} a_n &= \lim_{n \rightarrow \infty} n \sin(\frac{1}{n}) \\ &= \lim_{n \rightarrow \infty} \frac{\sin(\frac{1}{n})}{(\frac{1}{n})} \\ &= \lim_{\frac{1}{n} \rightarrow 0} \frac{\sin(\frac{1}{n})}{(\frac{1}{n})} \\ &= 1 \end{aligned}$$

$\therefore$ This series is divergent.

* Find $\sum_{n=1}^{\infty} \frac{2^n - 4^{n+2}}{5^{n-1}}$

$$\frac{2^n - 4^{n+2}}{5^{n-1}} = 2 \left(\frac{2}{5}\right)^{n-1} - 4^3 \left(\frac{4}{5}\right)^{n-1}$$

Both $\sum_{n=1}^{\infty} \left(\frac{2}{5}\right)^{n-1}$ and $\sum_{n=1}^{\infty} \left(\frac{4}{5}\right)^{n-1}$ are convergent geometric series with common ratio $\frac{2}{5}$ and $\frac{4}{5}$ respectively.

Further $\sum_{n=1}^{\infty} \left(\frac{2}{5}\right)^{n-1} = \frac{1}{1-\frac{2}{5}} = \frac{5}{3}$

$$\sum_{n=1}^{\infty} \left(\frac{4}{5}\right)^{n-1} = \frac{1}{1-\frac{4}{5}} = 5$$

$$\sum_{n=1}^{\infty} r^{n-1} = \frac{1}{1-r}$$

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{2^n - 4^{n+2}}{5^{n-1}} &= 2 \sum_{n=1}^{\infty} \left(\frac{2}{5}\right)^{n-1} - 4^3 \sum_{n=1}^{\infty} \left(\frac{4}{5}\right)^{n-1} \\ &= 2 \cdot \frac{5}{3} - 64 \times 5 \\ &= \frac{10}{3} - 320 \\ &= \frac{-950}{3} \end{aligned}$$

* Show that $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent. $\leftarrow$ comparison test.

Let $a_n = \frac{1}{n^2}$

$$a_n = \frac{1}{n^2} = \frac{1}{n \cdot n} < \frac{1}{n} \cdot \frac{1}{n-1} \quad \forall n = 2, 3, 4, \dots$$

Let $b_n = \frac{1}{n} \cdot \frac{1}{n-1} = \frac{1}{n-1} - \frac{1}{n}$

$$\Rightarrow b_2 = \frac{1}{2} - \frac{1}{3}$$

$$b_3 = \frac{1}{3} - \frac{1}{4}$$

$$b_{n-1} = \frac{1}{n-1} - \frac{1}{n}$$

$$b_n = \frac{1}{n} - \frac{1}{n+1}$$

$$\sum_{k=2}^n b_k = 1 - \frac{1}{n}$$

$$\therefore \sum_{n=1}^{\infty} b_n = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right) = 1$$

$\therefore \sum_{n=1}^{\infty} b_n$ is convergent.

$\therefore$ by the comparison test, $\sum_{n=1}^{\infty} a_n$ is convergent.

Show that $\sum_{n=1}^{\infty} \frac{2^n}{3^n(n+1)^2+9}$ is convergent.

$$\text{Let } a_n = \frac{2^n}{3^n(n+1)^2+9}$$

$$a_n = \frac{2^n}{3^n(n+1)^2+9} < \frac{2^n}{3^n(n+1)^2} < \frac{2^n}{3^n} = \left(\frac{2}{3}\right)^n = b_n$$

$$\text{Let } b_n = \left(\frac{2}{3}\right)^n \quad \forall n \in \mathbb{N}$$

$$\text{Then, } \sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n = \sum_{n=1}^{\infty} \left(\frac{2}{3}\right) \left(\frac{2}{3}\right)^{n-1} = \frac{2}{3} \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^{n-1}$$

This is geometric series with common ratio $r = \frac{2}{3} \in (1, 1)$

So, $\sum_{n=1}^{\infty} b_n$ is convergent.

$\therefore \sum_{n=1}^{\infty} a_n$ is convergent by the comparison test.

* Consider $\sum_{n=1}^{\infty} \frac{1}{5n-3}$

$$\frac{1}{5n-3} > \frac{1}{5n} \quad \forall n \in \mathbb{N}$$

$\underbrace{\hspace{1cm}}_{b_n} \qquad \underbrace{\hspace{1cm}}_{a_n}$

$$\text{Let } a_n = \frac{1}{5n} \text{ and } b_n = \frac{1}{5n-3}$$

Then $\sum_{n=1}^{\infty} a_n$ is divergent series.

$\therefore$ By the comparison test, $\sum_{n=1}^{\infty} b_n$ is divergent.

* Consider $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$ $\leftarrow$ check absolutely convergent.

$$\text{Let } a_n = \frac{(-1)^n}{n^2}, \quad \therefore |a_n| = \frac{1}{n^2} \quad \forall n \in \mathbb{N}$$

$$\text{Then } \sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} \frac{1}{n^2} \text{ is convergent.}$$

$\therefore \sum_{n=1}^{\infty} a_n$ is absolutely convergent.

* Consider $\sum_{n=1}^{\infty} \frac{n}{8n^3+9}$ $\leftarrow$ limit form of comparison test.

$$\text{Let } a_n = \frac{n}{8n^3+9}$$

$$\frac{n}{8n^3+9} < \frac{n}{8n^3} < \frac{1}{8n^2} < \frac{1}{n^2} = b_n$$

$$\text{Let } b_n = \frac{1}{n^2},$$

$$\text{Now, } L = \lim_{n \rightarrow \infty} \frac{a_n}{b_n}$$

$$L = \lim_{n \rightarrow \infty} \left(\frac{\frac{n}{8n^3+9}}{\frac{1}{n^2}} \right)$$

$$L = \lim_{n \rightarrow \infty} \left(\frac{n^3}{8n^3+9} \right)$$

$$L = \lim_{n \rightarrow \infty} \left(\frac{1}{8 + 9/n^3} \right) = \frac{1}{8}$$

$$\therefore 0 < \frac{1}{8} < \infty$$

Also we know, $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent.

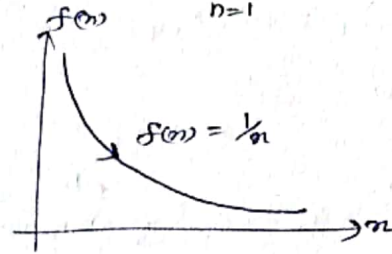
$\therefore$ By the limit form of comparison test;

$$\sum_{n=1}^{\infty} \frac{n}{8n^3+9} \text{ is convergent.}$$

* Use the integral test to show that $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent.

Let $a_n = \frac{1}{n}$ & $n \in \mathbb{N}$

Let $f(x) = \frac{1}{x}$ & $x > 0$



Then,

(i) $f(x) > 0$ & $x \in [1, \infty)$

(ii) $f(x)$ is decreasing on $[1, \infty)$

(iii) $f(x)$ is cts on $[1, \infty)$

Now, using the integral test,

$$\begin{aligned} \lim_{n \rightarrow \infty} \int_1^n f(x) dx &= \lim_{n \rightarrow \infty} \int_1^n \frac{1}{x} dx \\ &= \lim_{n \rightarrow \infty} \ln|x| \Big|_1^n \\ &= \lim_{n \rightarrow \infty} (\ln(n) - \ln(1)) \\ &= \infty \notin (0, \infty) \end{aligned}$$

$\therefore$ By the Integral test, $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent.

* Consider $\sum_{n=1}^{\infty} \frac{1}{n \ln n}$.

Here, $a_n = \frac{1}{n \ln n}$ & $n \in \mathbb{N}$

Let, $f(x) = \frac{1}{x \ln(x)}$ & $x > 1$

Then,

(i) $f(x) > 0$ & $x \in [2, \infty)$

(ii) $f(x)$ is decreasing

(iii) $f(x)$ is cts.

$$\begin{aligned} \text{Now, } \lim_{n \rightarrow \infty} \int_1^n \frac{1}{x \ln x} dx &= \lim_{n \rightarrow \infty} \int_1^n \frac{1}{\ln x} d(\ln x) \\ &= \lim_{n \rightarrow \infty} \ln|\ln x| \Big|_1^n \\ &= \lim_{n \rightarrow \infty} (\ln(\ln n) - \ln(\ln(1))) \\ &= \infty \notin (0, \infty) \end{aligned}$$

$\therefore$ by the integral test, this is divergent.

Consider $\sum_{n=1}^{\infty} \frac{1}{n^p}$ where $p \in \mathbb{R}$. $\leftarrow$ This is known as p-series.

Case I :- when $p < 0$

$$\text{Let } a_n = \frac{1}{n^p}$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1}{n^p} = \infty \neq 0$$

$$\therefore \sum_{n=1}^{\infty} \frac{1}{n^p} \text{ is divergent.}$$

$$\text{Let } f(x) = \frac{1}{x^p} \text{ for } x > 0$$

It's clear that f is a non-negative decreasing continuous function on $[1, \infty)$

$$\lim_{n \rightarrow \infty} \int_1^n f(x) dx = \lim_{n \rightarrow \infty} \left[\int_1^n \frac{1}{x^p} dx \right] \quad \text{--- (1)}$$

Case II :- when $p = 0$.

$$\text{(1)} \Rightarrow \lim_{n \rightarrow \infty} \left[\int_1^n f(x) dx \right] = \lim_{n \rightarrow \infty} \int_1^n 1 dx$$

$$= \infty \notin (0, \infty)$$

$$\therefore \sum_{n=1}^{\infty} \frac{1}{n^p} \text{ is divergent.}$$

Case III :- when $0 < p < 1$

$$\begin{aligned} \text{(1)} \Rightarrow \lim_{n \rightarrow \infty} \int_1^n f(x) dx &= \lim_{n \rightarrow \infty} \int_1^n \frac{1}{x^p} dx \\ &= \lim_{n \rightarrow \infty} \left. \frac{x^{-p+1}}{-p+1} \right|_1^n \\ &= \lim_{n \rightarrow \infty} \left[\frac{n^{-p+1}}{(-p+1)} - \frac{1}{(-p+1)} \right] \\ &= \infty \notin (0, \infty) \end{aligned}$$

$$\therefore \sum_{n=1}^{\infty} \frac{1}{n^p} \text{ is divergent.}$$

Case IV :- when $p = 1$,

$$\begin{aligned} \text{(1)} \Rightarrow \lim_{n \rightarrow \infty} \int_1^n f(x) dx &= \lim_{n \rightarrow \infty} \int_1^n \frac{1}{x} dx \\ &= \lim_{n \rightarrow \infty} \ln|x| \Big|_1^n \\ &= \lim_{n \rightarrow \infty} (\ln|n| - \ln|1|) \\ &= \infty \notin (0, \infty) \end{aligned}$$

$$\therefore \sum_{n=1}^{\infty} \frac{1}{n^p} \text{ is divergent.}$$

Case V: when $p > 1$,

$$\textcircled{1} \Rightarrow \lim_{n \rightarrow \infty} \int_1^n \frac{1}{x^p} dx = \lim_{n \rightarrow \infty} \left(\frac{x^{-p+1}}{-p+1} \Big|_1^n \right)$$

$$= \frac{1}{1-p} \in (0, \infty)$$

$\therefore \sum_{n=1}^{\infty} \frac{1}{n^p}$ is convergent.

$\sum_{n=1}^{\infty} \frac{1}{n^p}$ > convergent if $p > 1$
> divergent if $0 \leq p \leq 1$

* Consider $\sum_{n=1}^{\infty} \frac{(-1)^n (n^2+3)}{n^6+5n^3+7}$

$$a_n = \frac{(-1)^n (n^2+3)}{n^6+5n^3+7} \quad \forall n \in \mathbb{N}$$

$$|a_n| = \frac{n^2+3}{n^6+5n^3+7} \quad \forall n \in \mathbb{N}$$

$$\text{Let } b_n = \frac{1}{n^4} \quad \forall n \in \mathbb{N}$$

$\sum_{n=1}^{\infty} \frac{1}{n^4}$ is convergent.

$$\text{Now, } L = \lim_{n \rightarrow \infty} \frac{a_n}{b_n}$$

$$= \lim_{n \rightarrow \infty} \frac{n^2+3}{\cancel{n^6+5n^3+7} \cdot \frac{1}{n^4}}$$

$$= \lim_{n \rightarrow \infty} \frac{n^6+3n^4}{n^6+5n^3+7}$$

$$= 1 \in (0, \infty)$$

$\therefore$ ~~by comparison~~ $\sum_{n=1}^{\infty} \frac{1}{n^4}$ is convergent as it is a p-series with $p=4 > 1$.

$\therefore$ by L.F.C.T, the series $\sum_{n=1}^{\infty} |a_n|$ is convergent.

$\therefore$ given series $\sum_{n=1}^{\infty} a_n$ is absolutely convergent.

$\therefore$ It is convergent.

Consider the series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1} 3^n}{n!}$ ← limit form of the ratio test.

$$\text{Let } a_n = \frac{(-1)^{n-1} 3^n}{n!}$$

$$a_{n+1} = \frac{(-1)^n 3^{n+1}}{(n+1)!}$$

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{(-1)^n 3^{n+1}}{(n+1)!} \cdot \frac{n!}{(-1)^{n-1} 3^n} \right| = \left| \frac{(-1) \cdot 3}{n+1} \right| = \frac{3}{n+1}$$

$$\rho = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left(\frac{3}{n+1} \right) = 0 //$$

∴ By the limit form of the ratio test, the series is absolutely convergent.

∴ Series is convergent.

* Consider $\sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{n!}$.

$$\text{Let } a_n = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{n!}$$

$$a_{n+1} = \frac{1 \cdot 3 \cdot 5 \cdots (2n+1)}{(n+1)!}$$

$$\begin{aligned} \left| \frac{a_{n+1}}{a_n} \right| &= \left| \frac{1 \cdot 3 \cdot 5 \cdots (2n+1)}{(n+1)!} \cdot \frac{n!}{1 \cdot 3 \cdot 5 \cdots (2n-1)} \right| \\ &= \left| \frac{2n+1}{n+1} \right| \\ &= \frac{2n+1}{n+1} // \end{aligned}$$

$$\rho = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{2n+1}{n+1} \right| = 2 // \quad \therefore \rho > 1$$

∴ By the limit form of the ratio test, the series is divergent.

* Consider $\sum_{n=1}^{\infty} (-1)^n$.

$$\text{Let } a_n = (-1)^n$$

$$a_{n+1} = (-1)^{n+1}$$

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{(-1)^{n+1}}{(-1)^n} \right| = |-1| = 1 //$$

$$\rho = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} 1 = 1 //$$

∴ By the limit form of the ratio test, the series cannot be applied. But we know this series is divergent.

* Consider $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ $\leftarrow$ Alternative Series Test - AST

$$\text{Let } a_n = \frac{(-1)^n}{n} \text{ for } n \in \mathbb{N}$$

$$c_n = \frac{1}{n} \text{ for } n \in \mathbb{N}$$

$$\text{Consider, } c_n - c_{n+1} = \frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)} > 0 \text{ for } n \in \mathbb{N}$$

$$\therefore c_n > c_{n+1}$$

$\therefore (c_n)$ is decreasing sequence.

$$\text{Also } \lim_{n \rightarrow \infty} c_n = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$\therefore$ By AST, the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ is convergent.

* Consider $\sum_{n=1}^{\infty} (-1)^{n+1} \left(\frac{2n+7}{8n-2} \right)^n$ $\leftarrow$ Root Test.

$$\text{Let } a_n = (-1)^{n+1} \left(\frac{2n+7}{8n-2} \right)^n$$

$$\Rightarrow |a_n| = \left| (-1)^{n+1} \left(\frac{2n+7}{8n-2} \right)^n \right| = \left(\frac{2n+7}{8n-2} \right)^n \text{ for } n \in \mathbb{N}$$

$$\Rightarrow |a_n|^{1/n} = \left(\frac{2n+7}{8n-2} \right)^{n \cdot \frac{1}{n}} = \frac{2n+7}{8n-2} > 0 \text{ for } n \in \mathbb{N}$$

$$\alpha = \limsup |a_n|^{1/n}$$

$$= \limsup \left(\frac{2n+7}{8n-2} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{2n+7}{8n-2}$$

$$= \frac{2}{8} = \frac{1}{4} < 1$$

$\therefore$ By the root test, this series is absolutely convergent.

$\therefore$ It is convergent.

Consider $\sum_{n=1}^{\infty} \left[\frac{3+(-1)^n}{3+2(-1)^n} \right]^n$

let $a_n = \left[\frac{3+(-1)^n}{3+2(-1)^n} \right]^n$

$$|a_n| = \left| \frac{3+(-1)^n}{3+2(-1)^n} \right|^n$$

$$|a_n|^{1/n} = \frac{3+(-1)^n}{3+2(-1)^n} \quad \forall n \in \mathbb{N}$$

$$= \begin{cases} 4/5 & \text{if } n \text{ is even} \\ 2 & \text{if } n \text{ is odd} \end{cases}$$

$$A = \{ |a_n|^{1/n}, |a_n|^{1/n}, |a_n|^{1/n}, \dots, |a_n|^{1/n}, \dots \}$$

$$= \{ 2, 4/5, 2, 4/5, \dots \}$$

$$\therefore \sup A = 2$$

$$\alpha = \limsup |a_n|^{1/n}$$

$$= \lim_{n \rightarrow \infty} \left[\sup_{k \geq n} |a_k|^{1/k} \right]$$

$$= \lim_{n \rightarrow \infty} 2$$

$$= 2 > 1$$

$\therefore$ By the Root test, the series is divergent.

* Consider $\sum_{n=0}^{\infty} x^n \leftarrow \text{Power Series.}$

$$\left[\sum_{n=0}^{\infty} a_n (x-c)^n \right]$$

$$\sum_{n=0}^{\infty} x^n = \sum_{n=0}^{\infty} 1 \cdot (x-0)^n$$

Here $a_n = 1 \quad \forall n = 0, 1, 2, \dots$

$$c = 0$$

$\therefore$ This is a power series centered at $c=0$.

$$\text{---} (\quad) \text{---}$$

$c=0$

When $x=1$,

$$\sum_{n=0}^{\infty} x^n = \sum_{n=0}^{\infty} 1^n = \sum_{n=0}^{\infty} 1 \quad \text{is divergent series.}$$

When $x=-1$,

$$\sum_{n=0}^{\infty} x^n = \sum_{n=0}^{\infty} (-1)^n \quad \text{is divergent series.}$$

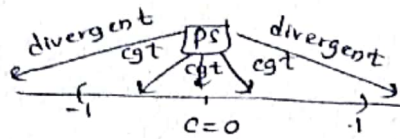
When $x=0$,

$$\sum_{n=0}^{\infty} x^n = \sum_{n=0}^{\infty} 0^n = 0 \quad \text{is convergent series.}$$

When $x = 1/2$,

$\sum_{n=0}^{\infty} x^n = \sum_{n=0}^{\infty} (1/2)^n$ is geometric series with common ratio $r = 1/2 < 1$ and so it is convergent.

$\therefore \sum_{n=0}^{\infty} x^n$ is convergent when $|x| < 1$ and is divergent $|x| \geq 1$.



* Consider $\sum_{n=0}^{\infty} (n+1)^2 (x-3)^n$

Here $a_n = (n+1)^2$ & $n \in \mathbb{N}$

$$c_n = 3$$

That is, this is a power series centered at $c=3$.

When $x=3$,

$$\sum_{n=0}^{\infty} (n+1)^2 (x-3)^n = \sum_{n=0}^{\infty} 0 \text{ is convergent.}$$

When $x=2$,

$$\begin{aligned} \sum_{n=0}^{\infty} (n+1)^2 (x-3)^n &= \sum_{n=0}^{\infty} (n+1)^2 (2-3)^n \\ &= \sum_{n=0}^{\infty} (n+1)^2 (-1)^n \\ &= \sum_{n=0}^{\infty} n^2 (-1)^{n-1} \text{ is divergent.} \end{aligned}$$

* Consider the power series: $\sum_{n=0}^{\infty} \frac{x^n}{n}$, Find R and $I.C.$

Here $a_n = 1/n$

$$\sum_{n=0}^{\infty} \frac{1}{n} (x-0)^n$$

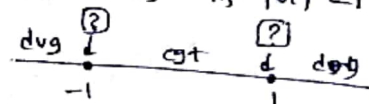
$$\begin{aligned} \lambda &= \limsup |a_n|^{1/n} \\ &= \limsup |1/n|^{1/n} \\ &= \lim_{n \rightarrow \infty} (1/n^{1/n}) \\ &= 1 \end{aligned}$$

$\therefore$ Radius of convergence,

$$R = \frac{1}{\lambda} = \frac{1}{1} = 1$$

Since $0 < R < \infty$, the power series is absolutely convergent, when $|x-c| < R$ and divergent when $|x-c| > R$

$\therefore$ The power series converges absolutely if $|x| < 1$



When $x=1$,

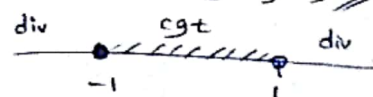
$$\sum_{n=1}^{\infty} \frac{x^n}{n} = \sum_{n=1}^{\infty} \frac{(1)^n}{n} = \frac{1}{n} \text{ is divergent.}$$

When $x=-1$,

$$\sum_{n=1}^{\infty} \frac{x^n}{n} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \text{ is divergent}$$

Interval of convergence,

$$\therefore I.C. \equiv [-1, 1)$$



Consider $\sum_{n=1}^{\infty} \frac{x^n}{n^2}$. Find R and IC .

Here $a_n = \frac{1}{n^2}$ & $n \in \mathbb{N}$

$c = 0$.

$$\sum_{n=1}^{\infty} \left(\frac{1}{n}\right)^2 (x-b)^n$$

$$\lambda = \limsup |a_n|^{1/n} = \limsup \left|\frac{1}{n^2}\right|^{1/n} = \lim_{n \rightarrow \infty} \frac{1}{n^{2/n}} = 1 \in (0, \infty).$$

$$R = \frac{1}{\lambda} = \frac{1}{1} = 1 //$$

$\therefore$ Radius of convergence, $R = 1 //$

Then, the power series converges if $|x| < 1$, i.e. $-1 < x < 1$

When $x = 1$,

$$\sum_{n=1}^{\infty} \frac{x^n}{n^2} = \sum_{n=1}^{\infty} \frac{1^n}{n^2} \text{ is convergent.}$$

When $x = -1$,

$$\sum_{n=1}^{\infty} \frac{x^n}{n^2} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \text{ is convergent.}$$

~~Therefore~~

$$\therefore IC = [-1, 1] // \leftarrow \text{Interval of convergence.}$$

* Consider $\sum_{n=1}^{\infty} n^n x^n$ Find R and IC .

Here $a_n = n^n$ & $n \in \mathbb{N}$, $c = 0$

$$\lambda = \limsup |a_n|^{1/n} = \limsup [n^n]^{1/n} = \lim_{n \rightarrow \infty} n = \infty.$$

$$\therefore R = \frac{1}{\lambda} = \frac{1}{\infty} = 0 //$$

$\therefore$ The power series converges only at $x = 0$. (center).

* $\sum_{n=0}^{\infty} \left[\frac{2+3n}{7+5n}\right]^n x^n$

Here $a_n = \left(\frac{2+3n}{7+5n}\right)^n$ & $n \in \mathbb{N}$

$c = 0$

$$\lambda = \limsup |a_n|^{1/n} = \limsup \left[\left(\frac{2+3n}{7+5n}\right)^n\right]^{1/n}$$

$$= \lim_{n \rightarrow \infty} \left(\frac{2+3n}{7+5n}\right)$$

$$= \frac{3}{5} //$$

$$\therefore R = \frac{1}{\lambda} = \frac{1}{3/5} = \frac{5}{3} //$$

Then, the power series converges if $|x| < \frac{5}{3}$.

i.e. $-\frac{5}{3} < x < \frac{5}{3}$

$$\left(\frac{2+7}{46}\right)^n \left(\frac{5}{3}\right)^n$$

when $x = \frac{5}{3}$,

$$\sum_{n=0}^{\infty} \left[\frac{2+5}{7+25/3}\right]^n \left(\frac{5}{3}\right)^n = \sum_{n=0}^{\infty} \left(\frac{5}{3}\right)^n \text{ is divergent.}$$

When $x = -\frac{5}{3}$,

$$\sum_{n=0}^{\infty} \left(\frac{2-5}{7-25/3}\right)^n \left(-\frac{5}{3}\right)^n \text{ is divergent.}$$

$$\therefore IC = \left(-\frac{5}{3}, \frac{5}{3}\right) //$$

* Consider $\sum_{n=0}^{\infty} \frac{n^3}{n!} x^n$. Find R and $\mathbb{I}C$.

$$a_n = \frac{n^3}{n!} > 0 \quad \text{if } n = 0, 1, 2, \dots$$

$$a_{n+1} = \frac{(n+1)^3}{(n+1)!}$$

Now, by theorem 2,

$$\begin{aligned} R &= \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{n^3}{n!} \cdot \frac{(n+1)!}{(n+1)^3} \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{n^3 (n+1)}{(n+1)^3} \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{n^3}{(n+1)^2} \right| \\ &= \infty \end{aligned}$$

$$\therefore \mathbb{I}C = R = (-\infty, \infty)$$

* Find the power series of $\frac{1}{1+x^2}$ for $|x| < 1$

We know that $\frac{1}{1-t} = 1 + t + t^2 + t^3 + \dots = \sum_{n=0}^{\infty} t^n \quad \text{--- (1)} \quad |t| < 1$
 ↑ power series of $\frac{1}{1-t}$.

Now, put $t = -x^2$ in (1),

$$\frac{1}{1-t} = \frac{1}{1-(-x^2)} = \frac{1}{1+x^2} = 1 + (-x^2) + (-x^2)^2 + (-x^2)^3 + \dots$$

$$\frac{1}{1+x^2} = 1 - x^2 + x^4 - x^6 + \dots$$

$$= \sum_{n=0}^{\infty} [-x^2]^n$$

$$\frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-1)^n x^{2n} \quad \text{--- (2)} \quad \text{if } |x| < 1$$

Hence, find the power series of $\tan^{-1}(x)$ for $|x| < 1$ Replace x by y

in (2), Then $\frac{1}{1+y^2} = \sum_{n=0}^{\infty} (-1)^n y^{2n} \quad \text{if } |y| < 1$

$$\Rightarrow \int_0^x \frac{1}{1+y^2} dy = \int_0^x \sum_{n=0}^{\infty} (-1)^n y^{2n} dy$$

$$[\tan^{-1} y]_0^x = \sum_{n=0}^{\infty} \int_0^x [(-1)^n y^{2n}] dy$$

$$(\tan^{-1} x - \tan^{-1} 0) = \sum_{n=0}^{\infty} (-1)^n \left[\frac{y^{2n+1}}{2n+1} \right]_0^x$$

$$\tan^{-1} x = \sum_{n=0}^{\infty} (-1)^n \left[\frac{x^{2n+1}}{2n+1} \right] \quad \text{if } |x| < 1$$

At $x = -1$, the power series becomes $\sum_{n=0}^{\infty} (-1)^n \frac{(-1)^{2n+1}}{(2n+1)}$

i.e. $\sum_{n=0}^{\infty} \frac{(-1)^{3n+1}}{(2n+1)} = - \sum_{n=0}^{\infty} \frac{(-1)^{2n}}{(2n+1)}$ is convergent by AST.

At $x = 1$, the power series becomes $\sum_{n=0}^{\infty} \frac{(-1)^n (1)^{2n+1}}{(2n+1)} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)}$

is also convergent by AST.

* Find the rational function of the power series $\sum_{n=1}^{\infty} n x^n$ for $|x| < 1$ and evaluate $\sum_{n=1}^{\infty} n/3^n$.

we want to find p and q s.t.

$$\sum_{n=1}^{\infty} n x^n = \frac{p(x)}{q(x)}$$

we know that, $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ for $|x| < 1$

Differentiating with respect to x gives,

$$-(1-x)^{-2} = 0 + \sum_{n=1}^{\infty} n x^{n-1}$$

$$-(1-x)^{-2} = \sum_{n=1}^{\infty} n x^{n-1}$$

$$-\frac{1}{(1-x)^2} \cdot x = \sum_{n=1}^{\infty} n x^{n-1} \cdot x$$

$$\therefore \sum_{n=1}^{\infty} n x^n = \frac{-x}{(1-x)^2} \text{ for } |x| < 1$$

* Let $f(x) = e^x$. Find Maclaurin series of f .

$$f(x) = e^x$$

$$f'(x) = e^x \rightarrow f'(0) = e^0 = 1$$

$$f''(x) = e^x \rightarrow f''(0) = e^0 = 1$$

...

$$f^{(n)}(x) = e^x \rightarrow f^{(n)}(0) = e^0 = 1$$

$\therefore$ The Maclaurin series of f is,

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = \sum_{n=0}^{\infty} \frac{1 \cdot x^n}{n!} = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$\therefore a_n = \frac{1}{n!}$$

Radius of convergence,

$$R = \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{1}{n!} \cdot \frac{(n+1)!}{1} \right|$$

$$= \lim_{n \rightarrow \infty} (n+1)$$

$$\therefore R = \infty$$

$$\therefore IC = (-\infty, \infty)$$

* Prove that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2} = 0$

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = L$$

Let $f(x,y) = \frac{x^2 y}{x^2 + y^2}$ and $L = 0$

$x^2 + y^2 = 0$ if and only if both $x = 0$ and $y = 0$.

$$\text{Domain of } f = D(f) = \{(x,y) \in \mathbb{R}^2 \mid (x,y) \neq (0,0)\} \\ = \mathbb{R}^2 - \{0,0\}$$

Take any $\epsilon > 0$, consider $|f(x,y) - L|$,

$$|f(x,y) - L| = \left| \frac{x^2 y}{x^2 + y^2} - 0 \right|$$

$$= \left| \frac{x^2 y}{x^2 + y^2} \right|$$

$$= \frac{x^2}{x^2 + y^2} |y|$$

$$\leq |y| \quad \text{since} \quad \frac{x^2}{x^2 + y^2} \leq 1 \quad \forall (x,y) \neq (0,0)$$

$$\leq \sqrt{x^2 + y^2} \quad \leftarrow y^2 \leq x^2 + y^2$$

$$= \sqrt{(x-0)^2 + (y-0)^2}$$

$$< \epsilon \quad \text{whenever} \quad \sqrt{(x-0)^2 + (y-0)^2} < \epsilon$$

we may take $\delta(\epsilon) = \epsilon$,

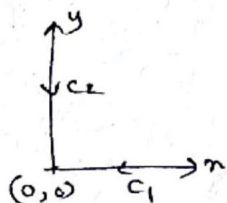
$$\therefore |f(x,y) - 0| < \epsilon \quad \text{whenever} \quad 0 < \sqrt{(x-0)^2 + (y-0)^2} < \delta(\epsilon)$$

$$\therefore \lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0.$$

* Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}$ does not exist.

Let $f(x,y) = \frac{x^2 - y^2}{x^2 + y^2}$

$$D(f) = \mathbb{R}^2 - \{0,0\}$$



Along c_1 : $x > 0$ and $y = 0$.

$$f(x,y) = \frac{x^2 - 0}{x^2 + 0} = \frac{x^2}{x^2} = 1$$

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \lim_{(x,y) \rightarrow (0,0)} 1 = 1 \quad \text{--- (1)}$$

Along c_2 : $y > 0$ and $x = 0$.

$$f(x,y) = \frac{0 - y^2}{0 + y^2} = -\frac{y^2}{y^2} = -1$$

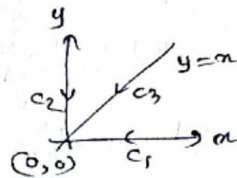
$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \lim_{(x,y) \rightarrow (0,0)} (-1) = -1 \quad \text{--- (2)}$$

$\therefore$ By (1) and (2), we conclude that the limit does not exist.

Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2+y^2}$ does not exist.

Let $f(x,y) = \frac{xy}{x^2+y^2}$

$D(f) = \mathbb{R}^2 - \{(0,0)\}$



along c_1 : $x > 0, y = 0$

$$f(x,y) = \frac{x \times 0}{x^2 + 0} = 0 //$$

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \lim_{(x,y) \rightarrow (0,0)} 0 = 0 // \text{--- (1)}$$

along c_2 : $y > 0, x = 0$

$$f(x,y) = \frac{0 \times y}{0 + y^2} = 0 //$$

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \lim_{(x,y) \rightarrow (0,0)} 0 = 0 // \text{--- (2)}$$

along c_3 : $y = x$ and $x > 0$

$$f(x,y) = \frac{x \cdot x}{x^2 + x^2} = \frac{x^2}{2x^2} = \frac{1}{2} //$$

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \lim_{(x,y) \rightarrow (0,0)} \frac{1}{2} = \frac{1}{2} // \text{--- (3)}$$

$\therefore$ By (1), (2) and (3), we conclude that limit does not exist. //

* Let $f(x,y) = \begin{cases} \frac{x^2y}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 3 & \text{if } (x,y) = (0,0) \end{cases}$ Is f continuous at $(0,0)$?

f is clearly defined at $(0,0)$

From a previous example, we know that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2y}{x^2+y^2} = 0$

But $f(0,0) = 3$.

$\therefore f$ is not continuous at $(0,0)$. //

* Let $z = e^x \sin y$, $x = t^2$ and $y = \sin t$. Find $\frac{dz}{dt}$.

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

$$\frac{\partial z}{\partial x} = e^x \cdot 0 + \sin y e^x = e^x \sin y.$$

$$\frac{dx}{dt} = 2t$$

$$\frac{\partial z}{\partial y} = e^x \cos y + \sin y \cdot 0 = e^x \cos y.$$

$$\frac{dy}{dt} = \cos t$$

$$\frac{dz}{dt} = e^x \sin y \cdot 2t + e^x \cos y \cdot \cos t$$

$$\frac{dz}{dt} = e^x (2t \sin y + \cos y \cos t), \text{ or}$$

$$\frac{dz}{dt} = e^{t^2} (2t \sin(\sin t) + \cos(\sin t) \cos t) //$$

* Let $z = e^x \sin y$, $x = st^2$, $y = s^2t$. Find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$.

$$\frac{\partial z}{\partial x} = e^x \cdot 0 + \sin y \cdot e^x = e^x \sin y //$$

$$\frac{\partial x}{\partial s} = s \cdot 0 + t^2 \times 1 = t^2 //$$

$$\frac{\partial z}{\partial y} = e^x \cos y + \sin y \cdot 0 = e^x \cos y //$$

$$\frac{\partial y}{\partial s} = s^2 \cdot 0 + t(2s) = 2st //$$

$$\begin{aligned} \frac{\partial z}{\partial s} &= e^x \sin y \cdot t^2 + e^x \cos y \cdot 2st \\ &= e^x (\sin y \cdot t^2 + 2st \cdot \cos y) \\ &= e^{st^2} (t^2 \sin(s^2t) + 2st \cos(s^2t)) // \end{aligned}$$

$$\frac{\partial x}{\partial t} = s \cdot 2t + t^2 \cdot 0 = 2st //$$

$$\frac{\partial y}{\partial t} = s^2 \cdot 1 + t \cdot 0 = s^2 //$$

$$\begin{aligned} \frac{\partial z}{\partial t} &= e^x \sin y \cdot 2st + e^x \cos y \cdot s^2 \\ &= s e^x (2t \sin y + s \cos y) \\ &= s e^{st^2} (2t \sin(s^2t) + s \cos(s^2t)) // \end{aligned}$$

* Let $f(x, y) = x^3 + 3xy^2 - 3x^2 - 3y^2 + 4$. Find and classify the critical points of f .

$$f_x = 3x^2 + 3y^2 - 6x //$$

$$f_y = 6xy - 6y //$$

The critical points of f are given by $f_x = 0$ and $f_y = 0$.

$$\Rightarrow 3x^2 + 3y^2 - 6x = 0$$

$$x^2 + y^2 - 2x = 0$$

$$\Rightarrow 6xy - 6y = 0$$

$$xy - y = 0$$

$$(x-1)y = 0 \begin{cases} x-1=0 \\ y=0 \end{cases}$$

Case I: $x^2 + y^2 - 2x = 0$ and $\boxed{y=0}$

$$x^2 - 2x = 0$$

$$(x-2)x = 0$$

$$\boxed{x=0} \text{ or } \boxed{x=2}$$

Then the critical points of f are $(2, 0)$, $(0, 0)$.

Case II : $x^2 + y^2 - 2x = 0$ and $x=1$

$$(1)^2 + y^2 - 2(1) = 0$$

$$1 + y^2 - 2 = 0$$

$$y^2 - 1 = 0$$

$$y = 1 \text{ or } y = -1$$

∴ Then the critical points of f are $(1, 1)$, $(1, -1)$.

∴ All the critical points of f are :

$$(2, 0), (0, 0), (1, 1), (1, -1)$$

To classify these points, we need at these points,

$$A = f_{xx} = \frac{\partial}{\partial x} (3x^2 + 3y^2 - 6x)$$

$$A = f_{xx} = 6x - 6$$

$$A = f_{xx} = 6(x-1)$$

$$B = f_{xy} = \frac{\partial}{\partial x} (6xy - 6y)$$

$$B = f_{xy} = 6y$$

$$C = f_{yy} = \frac{\partial}{\partial y} (6xy - 6y)$$

$$C = f_{yy} = 6x - 6$$

$$C = f_{yy} = 6(x-1)$$

Critical points.	A	B	C	$\Delta = AC - B^2$	Nature of critical points.
$(0, 0)$	-6	0	-6	$36 > 0$	Local maximum
$(2, 0)$	6	0	6	$36 > 0$	Local minimum
$(1, 1)$	0	6	0	$-36 < 0$	Saddle point
$(1, -1)$	0	-6	0	$-36 < 0$	Saddle point

* $\Delta > 0$, $A > 0$ ← local minimum.

$\Delta > 0$, $A < 0$ ← local maximum

$\Delta < 0$ ← saddle point.

* Find the minimum value of $f(x,y,z) = x^2 + y^2 + z^2$ subject to $2x + 3y - z = 1$

$$\text{Let } g = 2x + 3y - z - 1 = 0$$

$$2x + 3y - z = 1$$

$$y=0, z=0 \Rightarrow x = \frac{1}{2} \rightarrow \left(\frac{1}{2}, 0, 0\right)$$

$$x=0, z=0 \Rightarrow y = \frac{1}{3} \rightarrow \left(0, \frac{1}{3}, 0\right)$$

$$x=0, y=0 \Rightarrow z = -1 \rightarrow (0, 0, -1)$$

Let $F = f + \lambda g$ where $\lambda \neq 0$

$$F = (x^2 + y^2 + z^2) + \lambda (2x + 3y - z - 1)$$

$$F_x = 2x + 2\lambda$$

$$F_y = 2y + 3\lambda$$

$$F_z = 2z - \lambda$$

$$F_\lambda = (2x + 3y - z - 1)$$

now by Lagrange's multipliers at minimum of f , we have:

$$F_x = 2x + 2\lambda = 0 \quad \text{--- (1)}$$

$$F_y = 2y + 3\lambda = 0 \quad \text{--- (2)}$$

$$F_z = 2z - \lambda = 0 \quad \text{--- (3)}$$

$$F_\lambda = (2x + 3y - z - 1) = 0 \quad \text{--- (4)}$$

$$\text{(1)} \Rightarrow x = -\lambda //$$

$$\text{(2)} \Rightarrow y = -\frac{3}{2}\lambda //$$

$$\text{(3)} \Rightarrow z = \frac{1}{2}\lambda //$$

$$\text{(4)} \Rightarrow 2(-\lambda) + 3\left(-\frac{3}{2}\lambda\right) - \left(\frac{1}{2}\lambda\right) - 1 = 0$$

$$-2\lambda - \frac{9\lambda}{2} - \frac{\lambda}{2} - 1 = 0$$

$$-7\lambda = 1$$

$$\lambda = -\frac{1}{7} //$$

$$\therefore \text{ (4)} \Rightarrow \therefore x = \frac{1}{7} //$$

$$y = \frac{3}{14} //$$

$$z = -\frac{1}{14} //$$

$\therefore$ Minimum of f must occur at $\left(\frac{1}{7}, \frac{3}{14}, -\frac{1}{14}\right) //$

$\therefore$ The minimum of f =

$$f_{\min} = \left(\frac{1}{7}\right)^2 + \left(\frac{3}{14}\right)^2 + \left(-\frac{1}{14}\right)^2$$

$$= \frac{1}{49} + \frac{9}{196} + \frac{1}{196}$$

$$= \frac{14}{196}$$

$$= 0.071 //$$